



Dr. Tayeb Brahimi

Effat College of Engineering

Sciences, Technology, Engineering, and Mathematics Unit

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Google Profile: <https://scholar.google.com/citations?user=InKmJHYAAAAJ&hl=en>

Scopus Profile: <https://www.scopus.com/authid/detail.uri?authorId=57193233561>

Research Gate: <https://www.researchgate.net/profile/Tayeb-Brahimi>

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LinkedIn Profile: <https://www.linkedin.com/in/tayeb-brahimi-446793257/>

Short Biography:

Dr. Tayeb Brahimi is a distinguished associate professor at Effat University, Jeddah, Saudi Arabia. He earned his Ph.D. from the renowned Ecole Polytechnique Montreal, combining academic rigor with practical wisdom in his research and teaching. His career spans roles as a Research Scientist at Bombardier Chair in Canada, as well as a Programmer Analyst, QA Engineer, and Consultant in California and Texas, where he has made substantial contributions to engineering, renewable energy, sustainable development, artificial intelligence, and modern engineering education.

With over 150 publications, Dr. Brahimi has significantly advanced the fields of engineering and education, particularly through the integration of artificial intelligence (AI) models into problem-solving and research. He has contributed to the development of pivotal codes like CANICE for aircraft icing predictions and CARDAAS for wind turbine aerodynamics, while also leveraging AI models to optimize engineering processes. He is recognized globally for his achievements, earning accolades such as the Queen Effat Award for Excellence in Teaching and Research, "Best Presentation" at IEEE and IEOM in Singapore, and the prestigious Senior Fellow distinction from the UK Professional Standards Framework for Teaching and Learning.

Beyond his research, Dr. Brahimi is a passionate educator, always seeking innovative ways to make complex concepts accessible to his students. He actively incorporates AI-based tools into his teaching to provide students with hands-on experience in solving real-world engineering challenges. He is dedicated to empowering the next generation of engineers by integrating cutting-edge technologies into his curriculum, ensuring they are well-prepared for the dynamic challenges of the 21st century. As a visionary speaker, he is also committed to fostering innovation in engineering education.



EDUCATION

Ph.D./Master in Applied Sciences (MASc) (1987/1992)

Dept of Mechanical Eng., Ecole Polytechnique Montreal, Canada

Thesis title: Aerodynamic Analysis of Darrieus Rotor in Turbulent Wind

Advanced Level of Intensive English Program, Certificate of Proficiency, 1984

Concordia Continuing Education, Concordia University, Montreal, Canada

Bachelor of Engineering in Mechanical Engineering, 1983

Ecole Polytechnique, Algeria

Senior Project title: Plastic Deformation in Materials Processing

Advanced Blackboard Management System, 2016

Jeddah, KSA

Five Star Professional Development, Continuing Education, 2011

Jeddah, KSA

ESL/EFL Teaching Ways and Equipment of English Language, 2004

Mc Graw Hill, KSA

Quality Assurance, WinRunner Certification, 2002

Mercury Interactive, Silicon Valley, CA, USA

Power Electronic, Electronic Data Interchange Certification, 1999

Concord, CA, USA

TrustedLink UNIX/NT Edition, Concept & Software Implementation, 1998

Dallas, TX, Concord, USA

EDI Commercial Software UNIX, NT, and MVS, 1998

Dallas, TX, Concord, USA

Quality Code Implementation, 1998

Jeppesen DataPlan, Los Gatos, CA, USA

Meteorology, Protective Systems, Instrumentation, and Certification, 1993

University of Kansas, USA



TEACHING EXPERIENCE

Associate Professor, Effat College of Engineering, Natural Science, Math and Technology Unit, August 2020 – Present

Effat College of Engineering

Assistant Professor, Effat College of Engineering, Natural Science, Math and Technology Uni, August 2013 – August 2020

Effat College of Engineering

SAT Counselor/Supervisor and Math Teacher, May 2013-August 2013

GIS, Jeddah, KSA

Research Scientist, Bombardier/Canadair Chair, August 1992

January 1998

Ecole Polytechnique Montreal, Canada

Research Assistant, Bombardier Chair, January 1987-1992

Ecole Polytechnique Montreal, Canada

COURSES TAUGHT

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| ✓ Research Methodology in Engineering | ✓ Principles of Engineering |
| ✓ Foundations of Business Research | ✓ Fundamentals of Flight |
| ✓ Research Methodology for Thesis I & II | ✓ Energy Systems |
| ✓ Research, Innovation and Intellectual Property | ✓ Green Engineering & Environmental Compliance |
| ✓ Foundation of Business Research | ✓ Physics |
| ✓ Senior Projects in Eng. I & II | ✓ Applied Mathematics |
| ✓ Computer Applications | ✓ Algebra and Trigonometry for Eng. Students |
| ✓ VAWT & HAWT Wind Turbines | ✓ Statistics for Engineers |
| ✓ Solar Energy Systems | ✓ Thermodynamics |
| ✓ Renewable Energy | ✓ Dynamic Stall |
| ✓ Fluid Mechanics | ✓ Heat and Mass Transfer |
| ✓ Numerical Analysis | ✓ Aerodynamics |



ADMINISTRATIVE EXPERIENCE

Learning & Technology Program Co-Chair, Sept. 2015- Present

Effat University, Jeddah, KSA

General Secretary, Effat College of Engineering, Natural Science, Math and Technology Unit, August 2019 – 2022

Effat University, Jeddah, KSA

Supervisor, Spaceport Project, Siemens Saudi Arabia, Sept. 2021- Jan. 2022

Effat University and Siemens Saudi Arabia, Jeddah, KSA

Quality Assurance Committee, Effat College of Engineering, Natural Science, Math and Technology Unit, August 2018 – Present

Effat University, Jeddah, KSA

Supervisor, Solar System Installation, Effat College of Engineering, Sept. 2017-Jan. 2018

Effat University, Jeddah, KSA

Group Leader, Integration of innovative Islamic heritage into the STEM, collaboration EU and the Foundation for Science, (FSTC) UK partnership 2014-2018

Effat University - UK

Curriculum Development, member, College of Engineering, Graduate and Undergraduate programs, August 2015 – Present

Effat University, Jeddah, KSA

Math Coordinator, Effat College of Engineering, Natural Science, Math and Technology Unit, August 2015 – Present

Effat University, Jeddah, KSA

Academic Counselor & SAT Supervisor 2010-2013

GIS, Jeddah, KSA

Group Leader, Aircraft Icing, Bombardier Chair, 1992 - 1998



INDUSTRY EXPERIENCE

QA Engineer, June 1998 to August 2002

Peregrine System, Concord, CA and Inovis Systems, Dallas, Texas, USA

Programmer Analyst, Jan. 1998 – June 1998

Jeppesen DataPlan, Los Gatos, CA, USA

Technical Consultant, May 2015 – August 2018

IONPARA Inc., Wind Turbines & Aeronautical Consulting, Montreal, Canada

Research Scientists, Aug. 1992 – Jan.1998

J.-A. Bombardier/Canadair, Canada

RESEARCH EXPERIENCE

Research: Renewable Energy, Artificial Intelligence in Renewable Energy, Sustainability, ICT for Sustainable Development, Mathematical modeling, Student Learning Outcomes, Engineering Education, Accreditation, Makerspace, testing and quality assurance for component-based software, knowledge management, data mining, bibliometric analysis.

August 1992 - Present

Program Co-Chair, Learning & Technology Conference, Effat University, From 2015 - Present

Supervisor, Student Research Forum, “I-Research”, Effat university, 2017 – Present

Guest Editor, 5G, and Beyond; Paving the way for 6G Procedia Computer Science - Scopus Jan-21-May 21

Guest Editor "Artificial intelligence and machine learning research: towards digital transformation at a global scale - Journal of Ambient Intelligence and Humanized Computing" - 2021

Research Scientists, Bombardier Chair, University Montreal, 1992-1998

Group leader for various state-of-the-art math models in engineering used to simulate aircraft atmospheric icing (CANICE code) (<https://www.iopara.ca/software>).

Research Associate, Ecole Polytechnique, University Montreal, 1987-1992

Group leader for the various state-of-the-art wind energy modeling (CARDAAS code) for VAWT designers (<https://www.iopara.ca/software>).

Research and Development, Effat University, 2013 - Present

- Artificial Intelligence and Machine Learning in the Wind Power Industry
- Wind Farm Layout Modeling and Optimization Using Genetic Algorithm
- Wave Energy Prediction in the Red Sea
- Modeling and Numerical Analysis of Atmospheric Water Harvesting
- Atmospheric Water Harvesting Technology: Modelling vs. Experimental Approach
- Atmospheric Icing on Wind turbines and Aircraft
- Aerodynamic Analysis and Performance Prediction of Wind Turbines
- Wind Speed Distribution Using Machine Learning
- Solar Radiation Forecasting in Saudi Arabia Using Machine Learning
- Wind Energy in Saudi Arabia Opportunities and Challenges
- Bibliometric Analysis and Data Mining Research and Analysis
- Machine Learning Applied to Childhood Obesity, ADHD, and other Diseases.
- Artificial Intelligence and Machine Learning Research: Digital Transformation
- Sustainable Development Goals.
- Coping with Covid-19 in HES
- Research and Trends in COVID-19 Vaccines Using VOSviewer
- Artificial Intelligence and Covid-19: A Bibliometric Analysis
- A Survey of AI Solutions in response to the COVID-19 Pandemic in Saudi Arabia
- A Meta-Analysis of AI Application for Tracking COVID-19: The Case of UAE
- Monitoring the operation of transmission line in a smart grid system through IoT
- Smartphone-Based Human Activity Recognition using Bagging and Boosting
- ICT Education for Sustainable Development: Saudi Institutions of Higher Learning
- Powering makerspace wirelessly: Opportunities and challenges
- Makerspace: A Novel Approach to Creative Learning
- The Impact of Accreditation on Student Outcomes
- Learning Outside the Classroom through MOOCs

PUBLICATIONS

Published more than 150 articles in scientific journals, international conferences, and internal technical reports on wind energy, wave energy, solar energy, machine learning, aircraft icing, data mining, and novel methodologies for engineering education. For details, see: <https://scholar.google.com/citations?user=InKmJHYAAAAJ&hl=en>

PRESENTATIONS

Top Conference Presentations:

- American Institute of Aeronautics and Astronautics (AIAA), <https://arc.aiaa.org/>
- International Symposia on Transport Phenomena, Dynamics, and Design of Rotating Machinery, Boston, USA
- Springer Proceedings in Complexity book series (SPCOM), Italy
- Canadian Wind Energy Association, Canada
- National Conference of the American Wind Energy Association, USA
- European Community Wind Energy Conference, Glasgow
- ASME International Computers in Engineering Conference and Exhibition, USA
- Annual Iroquois Fluids Conference, Casowasco, New-York, USA
- Annual Vertical Axis Wind Turbine Aerodynamics Seminar, Sandia National Laboratories, USA
- Asian Congress of Fluid Mechanics, Hong Kong
- Intersociety energy conversion engineering conference, USA
- International Thermal Energy Congress ITEC-93, Marrakesh, Morocco
- Rotary-Wing Aircraft In-Flight Icing/De-Icing Workshop, Montreal, Canada
- International Symposium on Heat and Mass Transfer in Energy Systems and Environmental Effects, Mexico
- Bombardier International Workshop on Aircraft Icing and Boundary-Layer Stability and Transition, Canada
- International Forum on Global Environment and Friendly Energy Technology, Japan
- Int. Icing Symposium, Montreal, Canada
- Canadian Aerodynamic Symposium of CASI, Montreal, Canada
- International Workshop on Atmospheric Icing on Structures, Canada
- International Conference on Fluid Engineering (JSME), Tokyo, Japan

Invited Speaker:

- Faculty of New Technologies of Information and Communication (NTIC) Department of Software Technologies and Information Systems (TLSI), Constantine, Algeria, 2023
- Global Engineering Education (EIOM), 2019
- Engineering your Energy, Effat University, Solar and Wind Energy, Effat University, 2018.

- Global Liberal Arts Alliance. Education Workshop. Science Pedagogy Workshop at the American College of Greece, Greece, 2018
- Wind Turbine Design for K12 for Architecture students, Effat University
- Gulf Education Conference and Exhibition (GEC), in cooperation with the University of Business and Technology, Jeddah, KSA, 2017.
- The International Conference on Engineering Education & Research (FICEER), 2015
- The Gulf Educational Conference, Cambridge, 2014
- Participated in the Public Debate on Energy, Government of Quebec, Canada.
- Invited speaker by the Japan Society of Mechanical Engineering, Japan Turbomachinery Association (Ishikawajima-Harima Heavy Industries, Tokyo), 1997
- SAT & IGCSE course requirements, Jeddah, KSA, 2010-2013
- Smart Search Using Google, Effat University, 2004
- One of my publications, entitled "Learning Outside the Classroom through MOOCs," suggested ways of integrating MOOCs within the curriculum. IOWA State University currently uses this paper as an Online Learning Toolbox for students taking the EDUC-507 course: Principles and Practices of Distance Education graduate course. <https://iastate.pressbooks.pub/onlinelearningtoolbox/?s=brahimi>

REVIEWER

- Frontiers
- Advances in Artificial Intelligence and Machine Learning
- IEEE Access
- Cogent Engineering
- Energies
- Clean Energy
- Journal of Science and Technology, Emerald
- Applied Thermal Engineering
- Journal of Water Process
- NASA Space Apps Challenge
- Taylor and Francis, RII Conference, UK
- Learning and Technology Conference, Effat University, KSA
- Int. Journal of Rotating Machinery
- Behaviour & Information Technology Journal, UK
- Taylor and Francis, RII Conference, UK
- Journal of Ambient Intelligence and Humanized Computing, UK
- Elsevier, Procedia Computer Science, UK
- Computer Communication & Collaboration (CCC), Canada
- The Int. Conference on Cloud Technology and Communication Engineering, China
- International Journal for Research and Development in Technology, India

- Engineering Education & Teaching Conference, Greece
- Journal of the International Institute of Islamic Thought and Civilization (ISTAC), UAE
- Journal of Physics: IOP Conference Series
- The International Conference on Information Science and Technology, China
- The 19th Int. Conf. on Academic Learning and Administration in Higher Education, KSA
- Learning and Technology Conference, Effat University, KSA
- Reviewer for IDB French Committee
- Reviewer for ABET-SSR for ECE, CS, and IS
- Scientific and Technical Committee & Editorial Review Board Member for the World Academy of Science, Engineering, and Technology
- Learning & Technology, Effat University
- A judge in NASA Space Apps Challenge, April 27-29, 2017, Effat University, 2017
- and many others ...

GRANTS AND FELLOWSHIPS

- Effat University Grant (Wind Speed Forecasting Using Artificial Neural Network, 2020)
- Effat University Grant (Solar Irradiation Forecasting Using Artificial Neural Network, 2020)
- Effat University Grant (Green Smart Library, 2018)
- Bombardier Chair Grant, Canada (Aircraft Icing Predication, 1992-1998)
- Bombardier Chair Grant, Canada (Development of New Technology Computational Methods for Future Aircraft Design, 1996)
- Bombardier Chair Grant, Defense Research Establishment Valcartier, Quebec (Atmospheric Icing, 1995)

AWARDS AND HONORS

2014-2015: Queen Effat Award of Excellence in Teaching for Effat University

2019-2020: Queen Effat Award for Excellence in Research for Effat University

2020-2021: Queen Effat Award for Excellence in Teaching for Effat University

2022: Best Presentation "Atmospheric Water Harvesting", IEEE, Singapore

2021: Best Presentation "Covid-19 Competition", IEOM, Singapore

2021: Senior Fellow (SFHEA) - UK Professional Standards Framework for Teaching and Learning Support in Higher Education - Fellowship reference: PR210397

- 2021:** First Place of the “Covid-19 Competition”, Singapore
- 2019:** Best Track Paper Award, IEOM, GCC, Riyadh
- 2018:** Best Poster Award, Smart Grid, Jeddah
- 2018:** Best Poster Award, I-Research, Effat University
- 2018:** Best Video Demo Award, I-Research, Effat University
- 2018-2021:** Faculty Performance Recognition (100% Research, 100% Teaching)
- 2015:** I-Research Award for CoE Research and Publications
- 2014-2015:** Queen Effat Award of Excellence in Teaching for Effat University
- 2014-2015:** Award of Excellence in Research for the CoE, Effat University
- 2013-2014:** Best Student Evaluation, CoE, Effat University
- 2014-2016:** L&T Appreciation Awards, Effat University
- 2001:** The Employee of the Month, Concord, CA, USA
- 2001:** The Employee of the Quarter, Concord, CA, USA
- 1999:** The Employee of the Year Award, Concord, CA, USA
- 1999:** Appreciation Award - Technical Publication, Dallas, TX, USA
- 1996:** AIAA Minta Martin Student Competition, USA
- 1993:** Best Student Paper CASI-93, Toronto, Canada
- 1993:** Best Poster Paper ITEC-93, Marrakesh, Morocco

THESIS SUPERVISIONS

Ph.D. Dissertation co-supervision:

- François Morency, Simulation of Aircraft De-Icing, University Montreal
- Pascal Tran, Aircraft In-Flight Icing, University Montreal

Master Dissertation co-supervision:

- Dan, Chocran, Icing on Wind Turbines Blades, University Montreal
- Misaa, Alkhyat, Wave Energy and Wave Converters, Effat University
- Rawan Asfour, Layout Optimization of HAWTs in Wind Farms, Effat University
- Sadeem Alkinani, Numerical Simulation and Analysis of Harvesting Atmospheric Water Using Porous Materials, Effat University

Bachelor's Projects

- Many projects for undergraduate students: Wind Turbines, Wind Turbine Icing, Repaneling for wind power prediction, Atmospheric Icing, Machine Learning.

PROFESSIONAL MEMBERSHIPS

- Canadian Wind Energy Association
- American Institute of Aeronautics and Astronautics
- Effat Environment Club
- Senior Fellow (SFHEA) - UK Professional Standards Framework for teaching and learning support in higher education.

CERTIFICATIONS AND FELLOWSHIPS

- Innovative Teaching with ChatGPT
- Generative AI for Educators & Teachers
- High Impact Research and Citation Award
- GPT Vision: Seeing the World through Generative AI
- Advanced Blackboard Management System
- ABET and NCAAA Accreditation
- Quality Assurance, WinRunner Certification, Mercury Interactive, Silicon Valley, CA, USA
- Power Electronic, Electronic Data Interchange Certification, Concord, CA, USA
- TrustedLink UNIX/NT Edition, Concept & Software Implementation, Dallas, TX, USA
- EDI Commercial Software UNIX, NT, and MVS, Dallas, TX, Concord, USA
- Meteorology, Protective Systems, Instrumentation, and Certification, University of Kansas, USA

LIST OF PUBLICATIONS

Google Profile: <https://scholar.google.com/citations?user=InKmJHYAAAAJ&hl=en>

Scopus Profile: <https://www.scopus.com/authid/detail.uri?authorId=57193233561>

Research Gate: <https://www.researchgate.net/profile/Tayeb-Brahimi>

ORCID: <https://orcid.org/0000-0002-1575-4943>

Top TEN PUBLICATIONS (for more publications see Google Scholar:

https://scholar.google.com/citations?hl=en&user=InKmJHYAAAAJ&view_op=list_works)

1. T. Brahimi and A. Sarirete, "Transforming Learning in STEAM: How AI Tools and Language Models Catalyze Educational Advancement," in Transformative Leadership and Sustainable Innovation in Education: Interdisciplinary Perspectives, S. Baroudi and M.D. Lytras, Eds. Leeds: Emerald Publishing Limited, 2024, pp. 39-58. <https://doi.org/10.1108/978-1-83753-536-120241004>
2. T. Brahimi, A. Sarirete, HJ Al-Lail "Innovation, leadership, and education: how Effat University is paving the way for vision 2030," in Transformative Leadership and Sustainable Innovation in Education: Interdisciplinary Perspectives, S. Baroudi and M.D. Lytras, Eds. Leeds: Emerald Publishing Limited, 2024, pp. 169-189. <https://doi.org/10.1108/978-1-83753-536-120241011>
3. T. Brahimi, A. Sarirete, "Bibliometric analysis of GCC healthcare digital transformation. A bibliometric analysis of GCC healthcare digital transformation. Book Chapter, In Digital Transformation in Healthcare in Post-COVID-19 Times, Elsevier Academic Press.
<https://www.sciencedirect.com/science/article/abs/pii/B9780323983532000034>
4. T. Brahimi, "Data Mining and Visualization of Space Technology Research Trends in the Arab World," in 2023 20th Learning and Technology Conference (L&T), Jeddah, Saudi Arabia, 2023, pp. 179-183. <https://ieeexplore.ieee.org/document/10092340>
5. S. Alkinani, MF El-Amin, T. Brahimi. "Numerical Modeling and Analysis of Harvesting Atmospheric Water Using Porous Materials," J. Separations, 9(11), 364, 2022, <https://doi.org/10.3390/separations9110364>
6. R. Asfour, T. Brahimi, MF El-Amin "Wind Farm Layout: Modeling and Optimization Using Genetic Algorithm". 2022 IOP Conference Series: Earth and Environmental Science 1008 012004. <https://iopscience.iop.org/article/10.1088/1755-1315/1008/1/012004>
7. A. Sarirete, Z. Balfagih, T. Brahimi, M. D. Lytras, and A. Visvizi, "Artificial Intelligence and Machine Learning Research: Towards Digital Transformation at a Global Scale," J. Ambient Intell. Humaniz. Comput., Apr. 2021, doi: 10.1007/s12652-021-03168-y. <https://link.springer.com/article/10.1007/s12652-021-03168-y>
8. T. Brahimi, I. Paraschivoiu. "Aerodynamic Analysis and Performance Prediction of VAWT and HAWT Using CARDAAV and Qblade Computer Codes". 2021 In Book: "Wind Farms - Increasing Penetration, Economic Dispatch and Operational Planning ", IntechOpen , UK, April, 2021. <https://www.intechopen.com/online-first/aerodynamic-analysis-and-performance-prediction-of-vawt-and-hawt-using-cardaav-and-qblade-computer-c>
9. T. Brahimi. "Using Artificial Intelligence to Predict Wind Speed for Energy Application in Saudi Arabia." Energies, 2019, 12(24), 4669. <https://doi.org/10.3390/en12244669>
10. T. Brahimi and A. Sarirete, " Learning Outside the Classroom Through MOOCs," Computers in Human Behavior, Volume 51, Part B, October 2015, Pages 604-609. <https://doi.org/10.1016/j.chb.2015.03.013>



REFERENCES

Dr. Michael B Bragg, Professor,
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